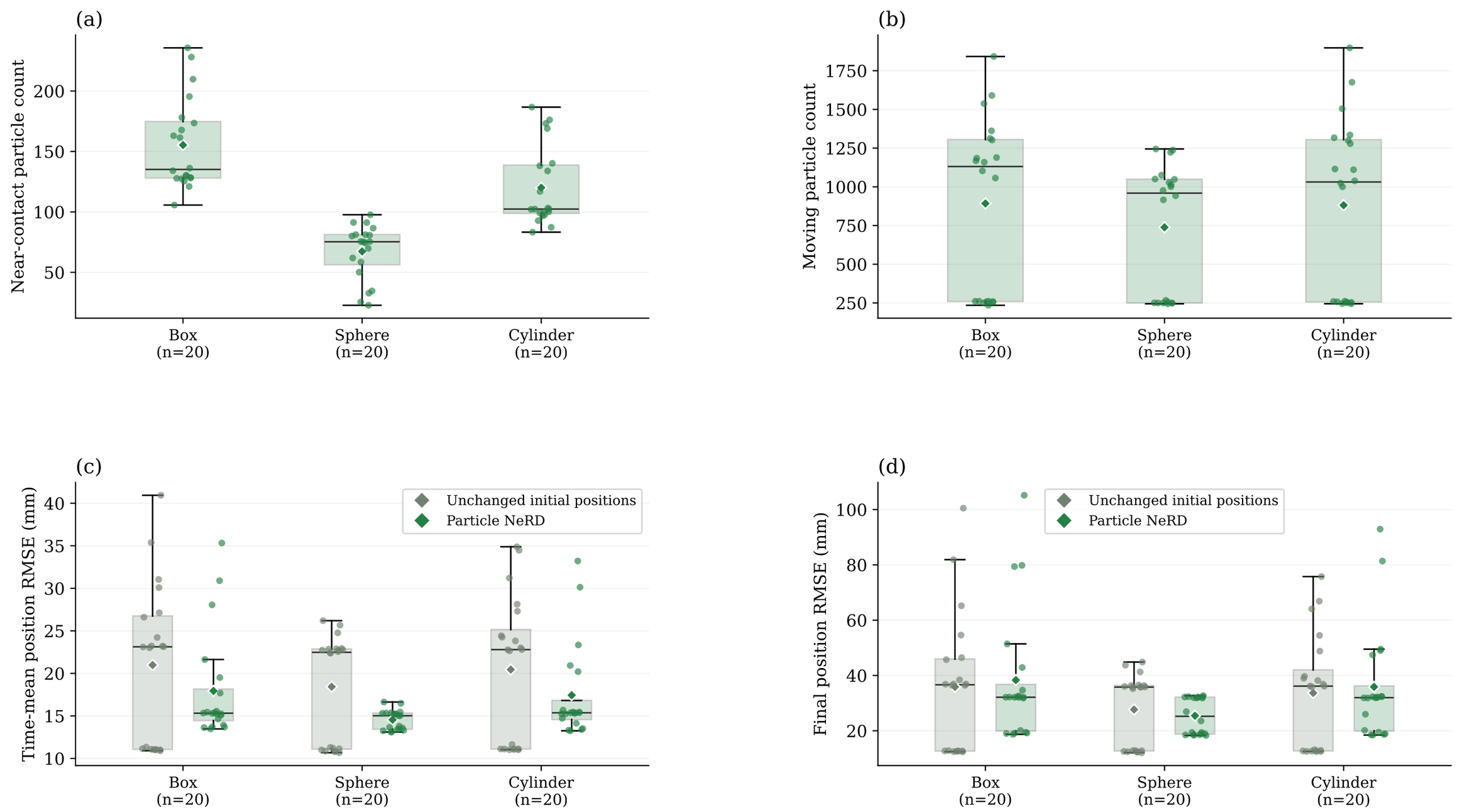

Figure 2: Frozen object-model checkpoint 9,000. (a,b) Time-mean reference populations within 0.05 m of the interacting-body surface and moving faster than 0.01 m/s. (c,d) Whole-bed time-mean and final position errors of NeRD versus retaining initial particle positions. The unchanged-position baseline is not another trained model. Each dot is one trajectory summary; boxes show the median and interquartile range, whiskers extend to 1.5 IQR, diamonds mark means. Twenty trajectories per shape unless labeled otherwise. Time means use available samples; empty particle regions are not assigned zero error.